

Priyanka Chatterjee

Backend Engineer · Node.js & TypeScript · REST APIs & Microservices · 9,300 ★ Open-Source Contributor · 3.5+ Years
Kolkata, West Bengal, India

+91 8617262536
priyankachatterjee9075@gmail.com
linkedin.com/in/priyanka-9075
github.com/Priyanka-Chatterjee-2000
priyanka-chatterjee-2000.github.io

SUMMARY

Backend Software Development Engineer with 3.5+ years of experience and expertise in **Node.js**, **TypeScript**, RESTful APIs, and backend **system design**. Led the **AWS** SDK for JavaScript v2 to v3 migration across 180+ call sites and 11+ **AWS** services in production **Node.js** Lambdas at Turbot HQ, cutting policy-code duplication ~60% while keeping the platform supported past **AWS** SDK v2 end-of-life.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Go, Python, SQL, Bash

Backend & APIs (Node.js focus): Node.js, Express.js, REST APIs, GraphQL, gRPC, Microservices, Distributed Systems, OAuth 2.0 (Auth0), JWT, Zod, Bun

Databases & SQL: PostgreSQL, MySQL, MongoDB, DuckDB, Redis (ElastiCache); jsonb, CTEs, query tuning, transactions, database migrations

Cloud & DevOps: AWS (Lambda, EC2, S3, SecurityHub, GuardDuty, ControlTower, Shield, X-Ray, EFS), Azure, GCP, Docker, Kubernetes (working knowledge), GitHub Actions (CI/CD)

Messaging & Event-Driven: Apache Kafka (consumer groups, DLQs), AWS SNS/SQS, AWS EventBridge, Azure Event Grid

Testing & Quality: Unit Testing, Integration Testing, End-to-End Testing, Jest, Mocha, Automated CI Testing, Code Review

Core CS: Data Structures & Algorithms, Object-Oriented Design, Backend Architecture, Low-Level & High-Level System Design, API Design

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Led the **AWS** SDK for JavaScript v2 to v3 migration across 180+ call sites and 11+ **AWS** services (SecurityHub, CloudWatch, ControlTower, Shield, Comprehend, Pinpoint, RAM, X-Ray, EC2, EFS) in production **Node.js** Lambdas, refactoring discovery, configuration-sync, tagging, and resource-cleanup workflows; reduced policy-code duplication ~60% via templating and shrank bundle size through modular imports, keeping the platform supported past **AWS** SDK v2 EOL and unblocking the **Node.js** runtime upgrade.
- Designed and developed a **REST API** microservice using the Gin framework as part of a greenfield build of Turbot Guardrails, a Preventive Security Posture Management (PSPM) SaaS product; secured endpoints with OAuth 2.0 authentication via Auth0, containerized with **Docker**, deployed to **AWS** EC2, and served as a gRPC client for several internal **microservices**.
- Core contributor across four open-source platform-engineering products (Steampipe, Powerpipe, Flowpipe, Tailpipe) — 120+ merged pull requests and 9,300+ GitHub stars — including engineering core throughput and memory optimizations for Tailpipe, lifting log-processing throughput 30% to 1,000+ logs per second via chunked, batched file reads and streaming writes, and eliminating out-of-memory failures.
- Optimized complex **PostgreSQL** queries powering customer compliance audits — rewriting IAM policy-evaluation queries with jsonb array elements and multi-stage CTEs — improving query performance 40% (cutting report generation from 3 minutes to 1.5 minutes across 10,000+ resources) and eliminating a class of false-positive findings.
- Built a **Kafka**-based, event-driven pipeline where consumers asynchronously processed policy-enforcement events and fired a webhook once a policy was successfully enforced against a resource; added a dead-letter queue (DLQ) to capture recurring failures for debugging, and unified **GitHub Actions** CI/CD release workflows across four production services.

PROJECTS

TODO / Tasks REST API (Apr 2026)

API-first **REST** backend in **TypeScript** with Bun, **Express.js** 5, and SQLite (bun:sqlite) + Zod + jose (JWT). JWT authentication (access, ID, and refresh tokens), RBAC, a prepared-statement CRUD service layer with crypto.randomUUID() IDs, schema-validated controllers with proper **REST** status codes (201/204/400/404), database migrations, and route-heatmap middleware.

EDUCATION

University of Engineering & Management (UEM) Kolkata, WB, IN

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design